Practical aspects of implementing the IRIDA system as a solution for One Health bioinformatics analyses

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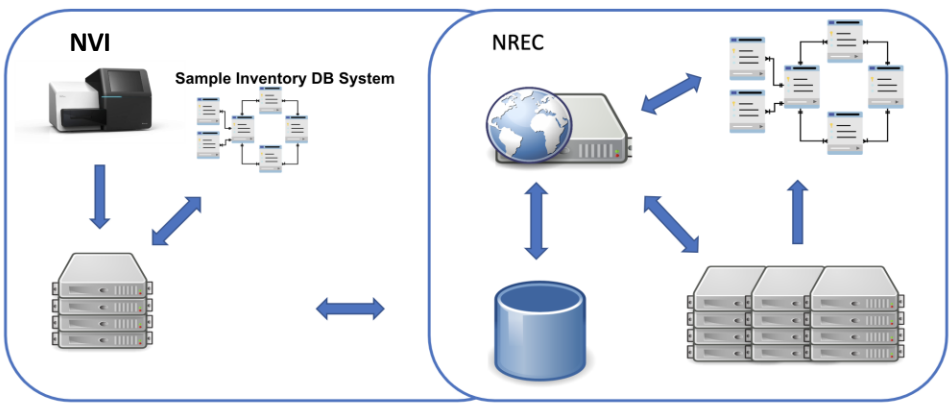
Norwegian Veterinary Institute (NVI) has installed Integrated Rapid Infectious Disease Analysis (IRIDA) platform. By doing so, NVI is enabling epidemiologists, clinical microbiologists and other molecular biologists to analyse whole genome sequences of bacteria and interpret the results through visualization tools. The IRIDA (irida.ca) platform has been developed by The Public Health Agency of Canada has together with other actors. IRIDA uses Galaxy for analyses and MySQL/PostgreSQL for storing user data and sample metadata. IRIDA comes with pipelines, visualization tools and a project based data management system that allows for fine grained data access control. At the NVI, we have set this system together with an academic cloud provider, and are actively adapting the setup and the pipelines to fit the needs of the institute.

IRIDA – benefits and tooling

- Open source, web based and user friendly analysis platform
- Uses Apache Tomcat, single WAR archive
- Needs MySQL/MariaDB database
- Drop into tomcat/webapps directory. Application autodeploys, and configures database after initial connection
- Communicates to galaxy through API key to share data and execute pipelines (galaxy workflows)

NVI setup

DNA squencing data will be transferred to NREC for analyses using IRIDA and the results will be sent back to NVI



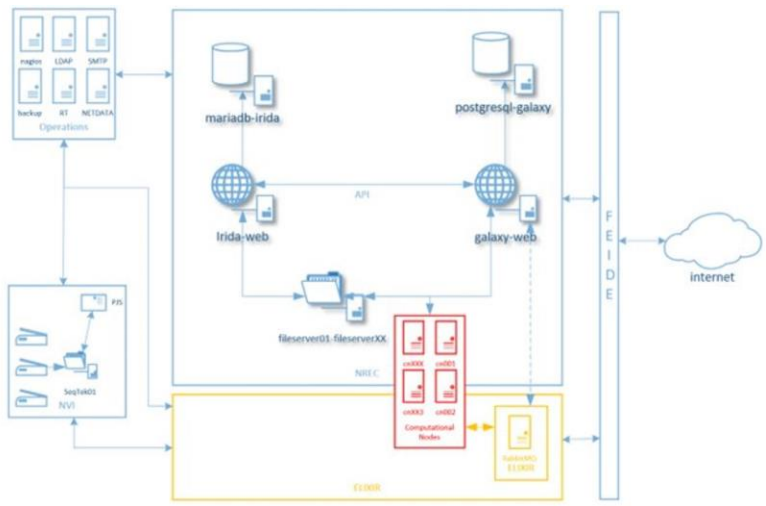
- **NREC:** Norwegian Research and Education Cloud
  - Flexible, inexpensive and sharing infrastructure knowledge
  - **OpenStack:** allows the creation of Virtual Machines out of pre-defined templates and pre-defined Operating Systems
  - Easy deployment of dynamic resources, with minimal intervention
- **Elixir Norway collaboration:** Dynamically spin up compute resources

Challenges we faced

- Multiple VMs to run Tomcat and DBMS
- Shared file system
- Moderate to advanced Galaxy admin knowledge
- Competence webservers, Java and Maven

Techinical Architecture

The image below shows the overall architecture of IRIDA installation at NREC

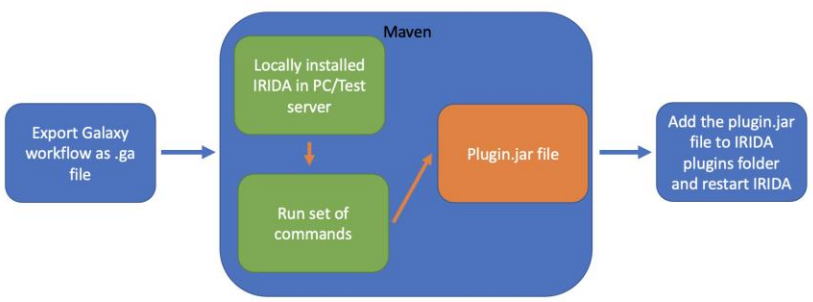


Elastic computation

We want to be able to spin up and down resources as needed, both for resource friendliness and financial reasons

- RabbitMQ - message queueing software, translates the job requirements from IRIDA/Galaxy to job requirements in ELIXIR-Norway
- PULSAR - instantiate jobs on remote servers without predefined use of servers

Pipelines



- Stitch the tools in Galaxy, create workflow and export it (.ga)
- Install IRIDA locally (Apache maven)
- Build the plugin by following IRIDA documentation (https://tinyurl.com/54hj8wdt) for pipeline development
- Modify the resulting XML and create java files based on the input and output of the pipeline
- Add the resulting .jar file into IRIDA and restart IRIDA (Tomcat)

Conclusion

- Successfully installed IRIDA/Galaxy in a cloud system which uses HPC
- Elastic computation allows efficient use of cloud infrastructure
- Added more pipelines from IRIDA pipeline repository
- Added our custom built pipelines as per our needs
- Future collaboration with ELIXIR-Norway for sharing cloud infrastructure

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